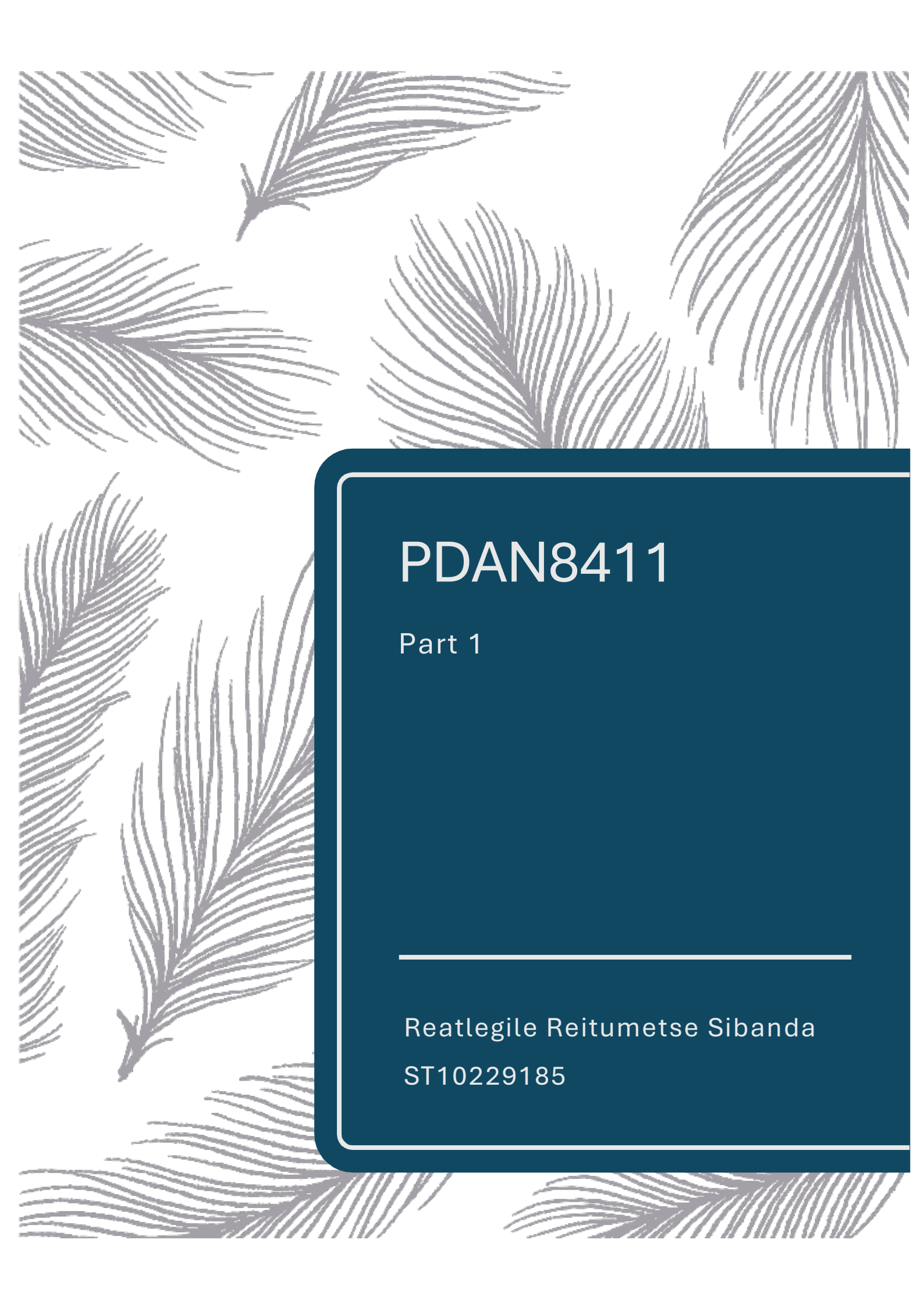




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Part 1

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Executive Summary

This report outlines a Linear Regression analysis aimed at estimating individual medical insurance costs based on demographic, lifestyle, and geographic factors. The analysis utilized the Medical Cost Personal Dataset, which includes data on age, sex, BMI, number of children, smoking habits, region, and medical expenses. Following a data quality assessment, one duplicate entry was eliminated, and no missing data points were found.

Exploratory Data Analysis revealed significant correlations between the predictors and medical expenses. Notably, smokers typically incurred much higher costs compared to non-smokers, while both age and BMI exhibited positive correlations with medical charges.

A foundational Multiple Linear Regression model yielded a Mean Absolute Error (MAE) of approximately \$4,177.05, a Root Mean Square Error (RMSE) of around \$5,956.34, and an R^2 value of approximately 0.8069 when evaluated on the test dataset.

An enhanced model was created by incorporating an interaction term between BMI and smoking status. This refined model achieved an MAE of about \$2,828.97, an RMSE of roughly \$4,572.81, and an R^2 of approximately 0.8862.

Consequently, the enhanced model demonstrated significantly superior predictive capabilities compared to the baseline model. Nevertheless, given that the dataset is based in the United States, this model should be regarded as a proof of concept and should not be directly applied to South African medical-aid pricing without appropriate South African data and additional validation.

Aim of the Study

The purpose of this study is to explore the potential for predicting individual medical insurance costs based on demographic, lifestyle, and geographic factors.

The specific aims include:

- Evaluating the appropriateness of the dataset for Linear Regression analysis.
- Analysing the quality and distribution of the data.
- Identifying significant correlations between customer attributes and medical expenses.
- Constructing a Multiple Linear Regression model utilizing Python.
- Assessing model efficacy through metrics such as MAE, MSE, RMSE, R^2 , and adjusted R^2 .
- Investigating whether the inclusion of an interaction term between BMI and smoking status enhances predictive accuracy.
- Interpreting the findings from a business standpoint while acknowledging the limitations inherent in the US-based dataset.

Data Cleaning and Exploratory Data Analysis

Initially, the dataset included 1,338 observations along with seven variables. A thorough missing-value analysis confirmed the absence of any missing data. One duplicate observation was identified and eliminated, resulting in a revised total of 1,337 observations for the modelling phase. The numerical variables underwent scrutiny for invalid values, and no clear negative or unrealistic observations were detected.

The Exploratory Data Analysis (EDA) revealed that medical charges were significantly right-skewed, indicating that a small fraction of customers faced substantially higher charges. Both age and BMI exhibited positive correlations with charges. The most significant visual disparity was found between smokers and non-smokers, with smokers generally incurring much higher charges.

The correlation analysis of numerical variables indicated that age had a correlation of approximately 0.30 with charges, BMI had a correlation of around 0.20, and the number of children showed a weak correlation of about 0.07.

These results informed the model development process. Age, BMI, and smoking status were retained as key predictors, while categorical variables were encoded through one-hot encoding. The strong correlation between BMI, smoking status, and charges also led to the consideration of a BMI-smoking interaction term in the refined model.

EDA Graphs

Graphs were shown on the notebook with findings examined

Model Training

The dataset was partitioned into training and testing subsets with an 80/20 ratio. The training subset comprised 1,069 observations, while the testing subset included 268 observations.

Multiple Linear Regression was chosen due to the continuous nature of the target variable, medical charges, alongside the availability of several predictor variables

Categorical variables were transformed into numerical indicator variables through one-hot encoding. The baseline model was developed utilizing age, BMI, number of children, sex, smoking status, and geographic region.

Statistical analysis was performed using p-values to assess the significance of individual predictors. Age, BMI, number of children, and smoking status were found to be statistically significant at the 5% level, whereas sex and the indicator for the Northwest region did not demonstrate statistical significance when controlling for the other predictors.

The model was further enhanced by incorporating an interaction term between BMI and smoking status. This decision was informed by exploratory data analysis (EDA), which revealed a notable disparity in charges between smokers and non-smokers, indicating that the relationship between BMI and charges might vary based on smoking status.

Model Evaluation and Interpretation

The baseline model recorded a Mean Absolute Error (MAE) of roughly \$4,177.05. This indicates that the average absolute discrepancy between the predicted and actual charges was about \$4,177.

The baseline Root Mean Square Error (RMSE) was around \$5,956.34. Given that RMSE places more emphasis on larger errors, this finding suggests that certain predictions exhibited significant deviations from the actual charges.

The baseline R^2 value was approximately 0.8069, signifying that nearly 80.7% of the variability in medical charges within the test dataset was accounted for by the predictors.

The enhanced model demonstrated superior performance across the primary evaluation metrics. The MAE improved from approximately \$4,177.05 to \$2,828.97, while the RMSE reduced from about \$5,956.34 to \$4,572.81. Additionally, the R^2 value rose from approximately 0.8069 to 0.8862.

This enhancement suggests that the interaction between BMI and smoking status contributed valuable predictive insights. Consequently, the improved model is favoured among those evaluated in this study.

Business Interpretation

The analysis reveals that customer attributes can yield valuable insights for predicting anticipated medical expenses. Smoking status is notably significant, as individuals who smoke in the dataset typically incur much higher medical costs compared to their non-smoking counterparts. Additionally, age and BMI serve as important predictive factors.

The organization must refrain from using the model as an automatic pricing tool without further validation. The pricing of medical aid encompasses regulatory, actuarial, ethical, and fairness considerations.

Importantly, the dataset analysed is based on US customers. The healthcare costs, medical schemes, population demographics, and regulatory conditions in South Africa are distinct from those represented in the dataset. Therefore, a South African application would require the acquisition of representative local data and comprehensive validation before implementation.

Conclusion

The research indicated that Multiple Linear Regression serves as an effective method for estimating medical insurance costs based on customer characteristics. The dataset was appropriate for a regression proof of concept, as it included a continuous numerical target alongside various potential predictors. Data cleaning revealed one duplicate entry and no missing values. Exploratory Data Analysis (EDA) uncovered significant relationships, notably the considerably higher charges linked to smokers.

The initial model accounted for roughly 80.7% of the variation in charges from the test set. By adding a BMI-smoking interaction, the enhanced model accounted for about 88.6% of the variation and significantly lowered both Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

These findings underscore the necessity of performing EDA prior to modelling and utilizing insights from EDA to guide model development. They also highlight the importance of employing multiple evaluation metrics rather than solely depending on R^2 .

While the enhanced model showed strong performance on the existing test data, it should not be deemed ready for production in the context of South African medical-aid pricing. The critical next step involves acquiring representative South African data and conducting additional validation, governance, and fairness evaluations before considering its operational deployment.